Import Data Using Transform Maps (Spreadsheet)

Phase 6 - Project Testing

1. Testing Overview

The **Project Testing Phase** is used to verify that the **"Import Data Using Transform Maps (Spreadsheet)"** solution works correctly according to the defined requirements.

The testing process verifies the complete data flow from the spreadsheet to the ServiceNow target table. It checks whether the data is imported correctly, fields are mapped properly, transformation rules work as expected, duplicate records are handled, and invalid data is identified.

The complete testing flow is:

Spreadsheet → Data Source → Import Set → Import Set Table → Transform Map → Validation → Target Table

2. Testing Objectives

The main objectives of the testing phase are:

- Verify successful spreadsheet data import.
- Verify Import Set configuration.
- Verify Import Set Table data.
- Verify Transform Map execution.
- Verify source-to-target field mapping.
- Verify data transformation.
- Verify Coalesce and duplicate handling.
- Verify mandatory field validation.
- Verify error handling.
- Verify new record creation.
- Verify existing record updates.
- Verify the accuracy of final target records.
- Ensure the solution meets the defined requirements.

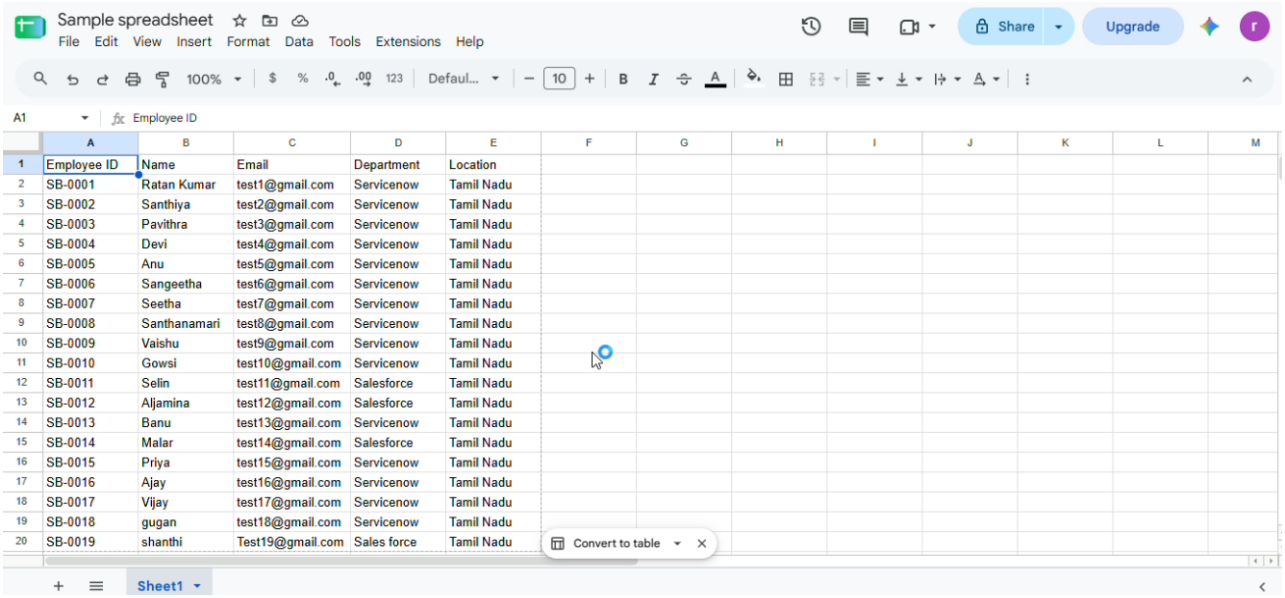
3. Testing Environment

Testing should be performed in a **ServiceNow development or test instance** instead of directly testing in production.

Required Test Components

- Test spreadsheet.

- ServiceNow test instance.
- Data Source.
- Import Set.
- Import Set Table.
- Transform Map.
- Target table.
- Test user account.
- Valid and invalid test data.



	A	B	C	D	E
1	Employee ID	Name	Email	Department	Location
2	SB-0001	Ratan Kumar	test1@gmail.com	Servicenow	Tamil Nadu
3	SB-0002	Santhiya	test2@gmail.com	Servicenow	Tamil Nadu
4	SB-0003	Pavithra	test3@gmail.com	Servicenow	Tamil Nadu
5	SB-0004	Devi	test4@gmail.com	Servicenow	Tamil Nadu
6	SB-0005	Anu	test5@gmail.com	Servicenow	Tamil Nadu
7	SB-0006	Sangeetha	test6@gmail.com	Servicenow	Tamil Nadu
8	SB-0007	Seetha	test7@gmail.com	Servicenow	Tamil Nadu
9	SB-0008	Santhanamari	test8@gmail.com	Servicenow	Tamil Nadu
10	SB-0009	Vaishu	test9@gmail.com	Servicenow	Tamil Nadu
11	SB-0010	Gowsi	test10@gmail.com	Servicenow	Tamil Nadu
12	SB-0011	Selin	test11@gmail.com	Salesforce	Tamil Nadu
13	SB-0012	Aljamina	test12@gmail.com	Salesforce	Tamil Nadu
14	SB-0013	Banu	test13@gmail.com	Servicenow	Tamil Nadu
15	SB-0014	Malar	test14@gmail.com	Salesforce	Tamil Nadu
16	SB-0015	Priya	test15@gmail.com	Servicenow	Tamil Nadu
17	SB-0016	Ajay	test16@gmail.com	Servicenow	Tamil Nadu
18	SB-0017	Vijay	test17@gmail.com	Servicenow	Tamil Nadu
19	SB-0018	gugan	test18@gmail.com	Servicenow	Tamil Nadu
20	SB-0019	shanthi	Test19@gmail.com	Sales force	Tamil Nadu

4. Test Data Preparation

Different types of spreadsheet data should be prepared for testing.

Valid Data

```
Employee ID: E001
Employee Name: Arun
Email: arun@example.com
Department: IT
Joining Date: 10-01-2025
```

Invalid Data

Examples:

- Missing Employee ID.
- Missing Employee Name.
- Invalid email address.
- Incorrect date format.
- Invalid department.
- Duplicate Employee ID.
- Empty spreadsheet row.

Using both valid and invalid data helps verify the reliability of the import process.

5. Functional Testing

Functional testing verifies whether each feature works according to the requirements.

Test Case 1 – Valid Spreadsheet Import

Test Objective: Verify that a valid spreadsheet can be imported successfully.

Steps:

1. Prepare a spreadsheet with valid records.
2. Upload the spreadsheet.

3. Load the data into the Import Set.
4. Execute the Transform Map.
5. Open the target table.

Expected Result:

All valid spreadsheet records should be successfully created in the target table.

Status: Pass/Fail

Test Case 2 – Import Set Table Verification

Test Objective: Verify that spreadsheet data is correctly stored in the Import Set Table.

Steps:

1. Upload the spreadsheet.
2. Run the import.
3. Open the Import Set Table.
4. Compare the imported values with the spreadsheet.

Expected Result:

The source data should appear correctly in the Import Set Table.

Test Case 3 – Field Mapping Verification

Test Objective: Verify that source fields are mapped to the correct target fields.

Example:

Spreadsheet Field	Target Field
Employee ID	Employee ID
Employee Name	Name
Email	Email
Department	Department
Joining Date	Date of Joining

Expected Result:

Each source value should appear in the corresponding target field.

6. Transform Map Testing

Test Case 4 – Transform Map Execution

Test Objective: Verify that the Transform Map executes successfully.

Steps:

1. Load spreadsheet data.
2. Open the Import Set.
3. Execute the Transform Map.
4. Review the transformation results.

Expected Result:

The Transform Map should execute without unexpected errors, and valid records should be transferred to the target table.

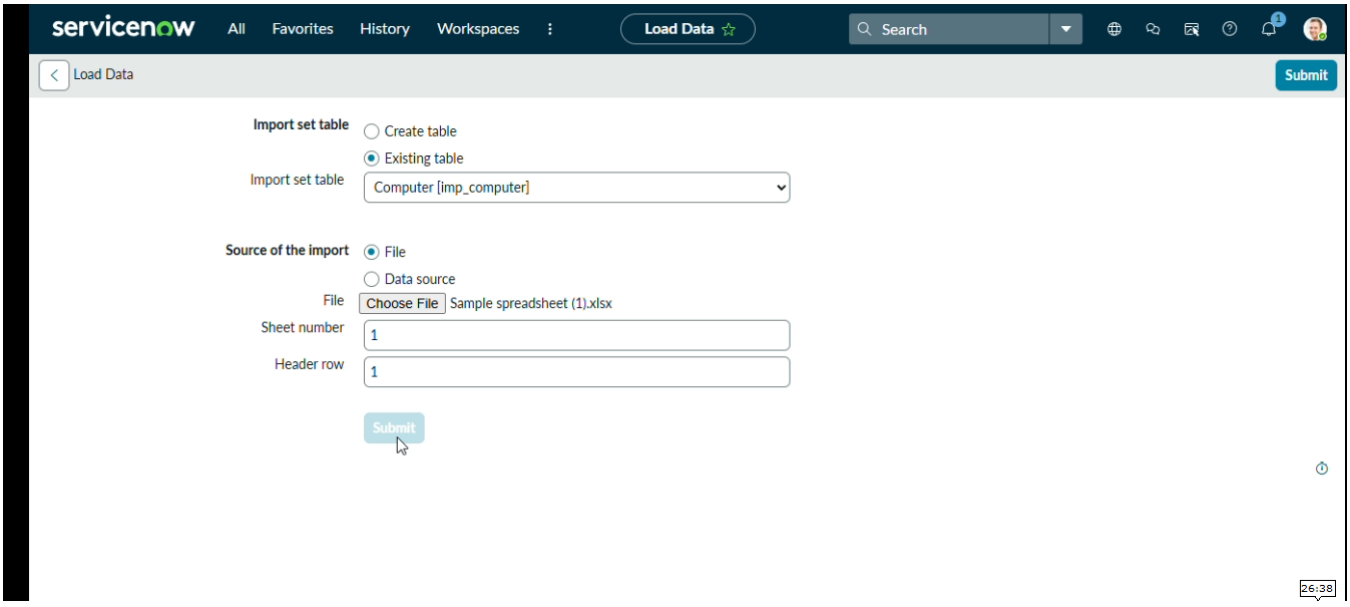
Test Case 5 – Data Transformation

Test Objective: Verify that transformation logic works correctly.

For example, if the source contains a specific date format and the target requires another format, the Transform Map should convert the value correctly.

Expected Result:

The target field should contain the correctly transformed value.



7. Coalesce and Duplicate Testing

Test Case 6 – New Record Creation

Input: Employee ID `E100`, which does not exist in the target table.

Expected Result:

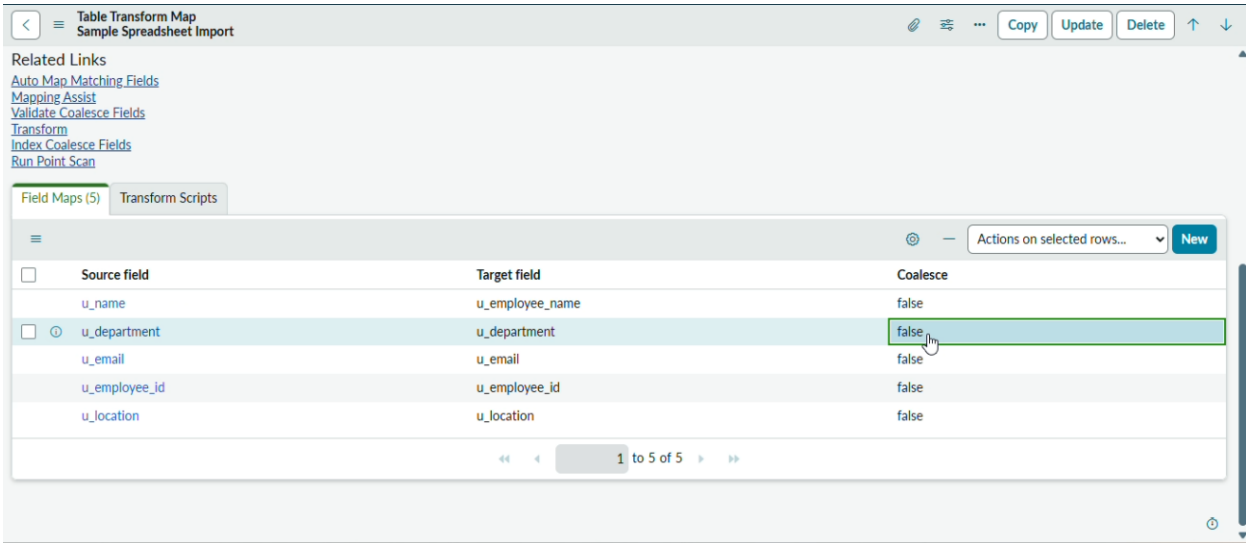
A new record should be created.

Test Case 7 – Existing Record Update

Input: Employee ID `E001`, which already exists.

Expected Result:

The existing record should be updated instead of creating another record.



Test Case 8 – Duplicate Record Handling

Input: Spreadsheet contains the same Employee ID more than once.

Expected Result:

The configured Coalesce and transformation logic should handle the duplicate according to the defined business requirement.

8. Data Validation Testing

Test Case 9 – Mandatory Field Validation

Input:

```
Employee ID: E005
Employee Name: Blank
Email: employee@example.com
```

Expected Result:

The record should be identified as invalid because the mandatory Employee Name is missing.

Test Case 10 – Invalid Email

Input:

```
Email: employee@invalid
```

Expected Result:

The invalid email should be identified according to the configured validation rules.

Test Case 11 – Invalid Date

Input:

```
Joining Date: ABC123
```

Expected Result:

The system should identify the invalid date value and prevent incorrect data from being inserted into the target field.

9. Error Handling Testing

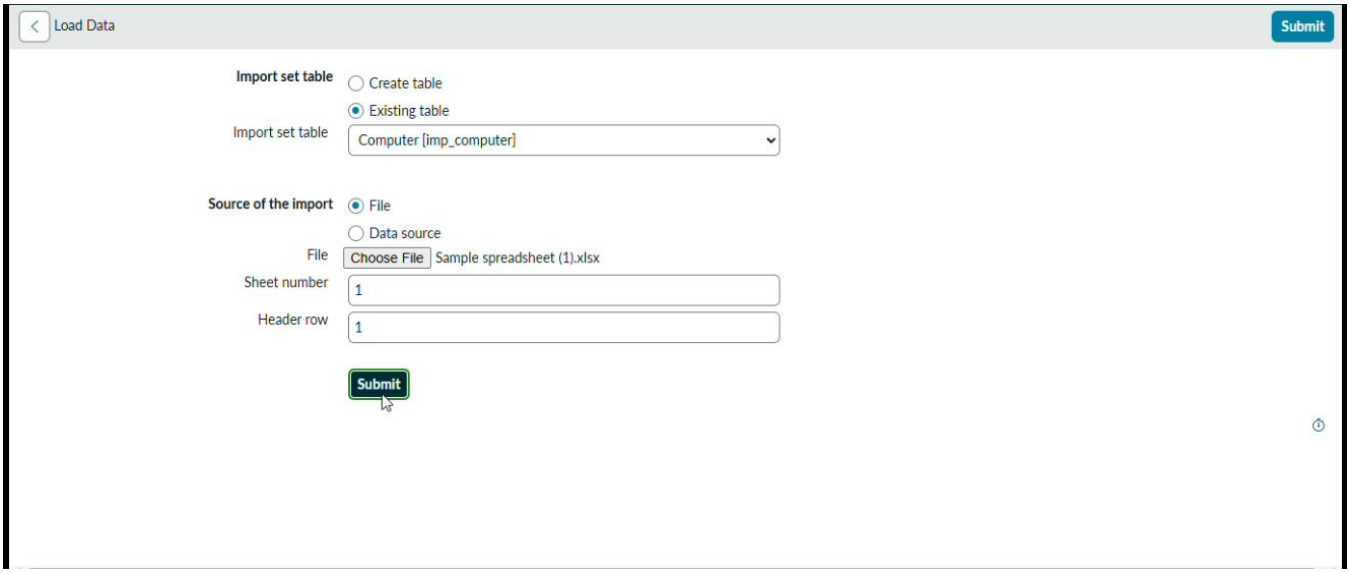
Error handling should be tested using incorrect or incomplete records.

Possible errors include:

- Missing mandatory values.
- Invalid field formats.
- Invalid reference values.
- Transformation failures.
- Incorrect field mapping.
- Duplicate records.

Expected Result

Errors should be identifiable through the import/transform results, allowing the user to correct the source data and perform the import again.



10. Negative Testing

Negative testing verifies how the system behaves when invalid data is provided.

Examples:

Test Input	Expected Result
Empty spreadsheet	Import should be handled appropriately
Missing Employee ID	Record should fail validation
Invalid email	Validation/error should occur
Invalid date	Transformation error should be identified
Invalid department	Record should be rejected or handled according to the defined rule

Duplicate ID	Coalesce logic should handle the record
Extra column	Unmapped field should not affect required mappings

11. Integration Testing

Integration testing verifies the complete data flow between all components.

Integration Flow

```
Spreadsheet
     ↓
Data Source
     ↓
Import Set
```

```
     ↓
Import Set Table
     ↓
Transform Map
     ↓
Field Mapping
     ↓
Transformation
     ↓
Validation
     ↓
Coalesce
     ↓
Target Table
```

Expected Result

Data should move successfully through all components without unexpected data loss or incorrect transformation.

12. User Acceptance Testing

User Acceptance Testing (UAT) ensures that the solution meets the actual business requirement.

The business user should verify:

- Spreadsheet data is imported correctly.
- Required fields contain correct values.
- New records are created correctly.
- Existing records are updated correctly.
- Duplicate handling works as expected.
- Invalid data is identified.
- Target table records meet business expectations.

The solution can be considered ready for deployment after successful user acceptance testing.

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13. Performance Testing

Performance testing verifies how the system behaves when importing a larger number of records.

Test scenarios can include:

- 10 records.
- 100 records.
- 500 records.
- 1,000+ records, depending on the project requirement.

The following should be monitored:

- Import processing time.
- Transform execution time.
- Number of successful records.
- Number of failed records.
- Overall system responsiveness.

14. Security Testing

Security testing ensures that only authorized users can perform the import operation.

The following should be verified:

- Unauthorized users cannot perform restricted imports.
- Authorized users have appropriate permissions.
- Target records follow existing access controls.
- Sensitive data is not exposed unnecessarily.
- Import-related access is configured appropriately.

15. Test Case Summary

Test ID	Test Scenario	Expected Result
TC01	Valid spreadsheet import	Records imported successfully
TC02	Import Set verification	Source data stored correctly
TC03	Field mapping	Correct target fields populated
TC04	Transform Map execution	Transformation completes successfully
TC05	Data transformation	Values transformed correctly
TC06	New record	New record created
TC07	Existing record	Existing record updated
TC08	Duplicate record	Duplicate handled correctly
TC09	Missing mandatory field	Validation/error triggered
TC10	Invalid email	Invalid data identified
TC11	Invalid date	Date error identified
TC12	Invalid reference value	Error/rejection handled
TC13	Large data import	Data processed successfully
TC14	Unauthorized access	Access restricted

16. Defect Management

If a test case fails, the issue should be recorded and analyzed.

The defect management process is:

Identify Defect → Record Defect → Analyze Cause → Fix Configuration → Retest → Verify → Close

Common defects may include:

- Incorrect field mapping.
- Wrong target field.
- Incorrect Coalesce configuration.
- Transformation script error.
- Invalid validation logic.
- Incorrect data format.
- Permission issues.

17. Retesting

After fixing a defect, the failed test case should be executed again.

For example:

Issue: Employee Name is mapped to the wrong target field.

Fix: Correct the Transform Map field mapping.

Retest: Import the spreadsheet again and verify the target field.

Expected Result: Employee Name should appear in the correct target field.

18. Regression Testing

Regression testing ensures that fixing one issue does not affect other working functionality.

For example, after changing the Coalesce configuration, the following should be tested again:

- New record creation.
- Existing record update.
- Duplicate handling.
- Field mapping.
- Transform Map execution.

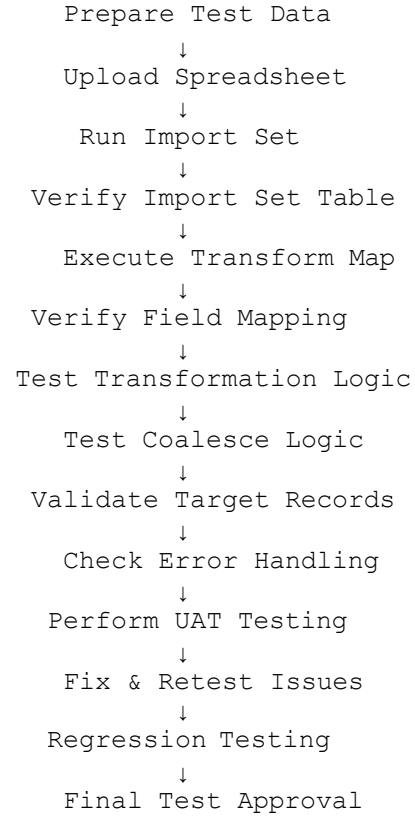
This ensures that the complete import process continues to work correctly.

19. Testing Completion Criteria

The testing phase can be considered complete when:

- All major functional test cases are executed.
- Field mappings are verified.
- Transform Map works correctly.
- Data transformation is successful.
- New records are created correctly.
- Existing records are updated correctly.
- Duplicate handling works correctly.
- Validation works correctly.
- Error handling is verified.
- Performance is acceptable.
- Security checks are completed.
- Critical defects are resolved.
- User Acceptance Testing is completed successfully.

20. Final Testing Flow



21. Final Project Testing Statement

The Project Testing Phase validates the complete “Import Data Using Transform Maps (Spreadsheet)” solution to ensure that it meets the defined functional and business requirements. Testing covers spreadsheet import, Import Set processing, field mapping, Transform Map execution, data transformation, Coalesce and duplicate handling, validation, error handling, performance, security, and user acceptance. Both positive and negative test scenarios are executed to identify defects and verify the reliability of the solution. Any identified issues are corrected and retested before final approval and deployment.